Early Experimental Studies of Cobalt-Ammines

*By George B. Kauffman**

IN JULY OF 1856 an important sixty-seven-page memoir, "Researches on the Ammonia-Cobalt Bases," by the American chemist Oliver Wolcott Gibbs (1822–1908) and the German-American chemist and mineralogist Frederick Augustus Genth (1820–1893) was accepted for publication as Article V of Volume IX in the Smithsonian Contributions to Knowledge series.[1] Although this monograph described in great detail a large number of the then-little-known cobalt-ammines, in this form it did not receive widespread circulation, especially abroad, and many citations of Gibbs' and Genth's work by workers active in the field refer to the reprinted version that appeared in Benjamin Silliman's *American Journal of Science* or to the abridged German translation that appeared in *Liebigs Annalen.*[2] Minor references to Gibbs' and Genth's work appear in historical portions of recent books specifically devoted to coordination chemistry, but they are ignored in most of today's standard histories of chemistry. Thus the purpose of this paper is to acquaint the general historian of science with the developments preceding and the circumstances surrounding Gibbs' and Genth's joint work, to relate it to the work of other nineteenth-century chemists, and to consider questions of priority in the preparation and analysis of crystallized cobalt-ammines, a contribution first made by Genth alone.

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[1] Wolcott Gibbs and Frederick Aug. Genth, *Researches on the Ammonia-Cobalt Bases* (Washington, D.C.: Smithsonian Institution, 1856).

[2] Wolcott Gibbs and F. A. Genth, "Researches on the Ammonia-Cobalt Bases," *American Journal of Science*, 1856, (2) *23*:235–265, 319–341; 1857, (2)*24*:86–107. "Ueber ammoniakalische Kobaltverbindungen; nach W. Gibbs and F. Genth," *Liebigs Annalen der Chemie,* 1857, *104*:150–174, 295–324. See, e.g., the citations of C. W. Blomstrand, *Die Chemie der Jetztzeit vom Standpunkte der electrochemischen Auffassung aus Berzelius Lehre entwickelt* (Heidelberg: Carl Winter's Universitätsbuchhandlung, 1869), pp. 281, 294; Sophus Mads Jørgensen's first article on coordination compounds, "Beiträge zur Chemie der Kobaltammoniakverbindungen. I. Ueber die Chloropurpureokobaltsalze," *Journal für praktische Chemie,* 1878, [2] *18*:209–247, pp. 209, 220, 229, 232, 238, 246; G. Vortmann and G. Magdeburg, "Ueber die Einwirkung der schwefligen Säure auf Kobaltammoniumsalze," *Berichte der Deutschen Chemischen Gesellschaft,* 1889, *22*:2630–2637, p. 2630. Also, in Jørgensen's last paper on metal-ammines he lists "mainly Genth, Gibbs, Cleve, F. Rose, Cossa, Vortmann, Blomstrand, and myself" as "those who had dealt experimentally more extensively with metal-ammonia salts" ("Zur Konstitution der Kobalt-, Chrom- und Rhodiumbasen. XI.Mitteilung," *Zeitschrift für anorganische Chemie,* 1899, *19*:109–157, p. 115). By the time that Alfred Werner had appeared on the scene (1893), the early experimental work of Gibbs and Genth (1856, 1857) had already been superseded by later studies by Jørgensen and others, so that Werner's references to Gibbs and Genth are few (e.g., Alfred Werner and Alfred Klein, "Beitrag zur Konstitution anorganischer Verbindungen. VI.Mitteilung. Über 1.6-Dichlorotetramminkobaltisalze (Chloropraseosalze)," *Z. anorg. Chem.,* 1897, *14*:28–41, p. 28; A. Werner, "Über 1.2-Dichloro-tetrammin-kobaltisalze (Ammoniakvioleosalze)," *Ber. Deutsch. Chem. Gesell.,* 1907, *40*:4817–4825, p. 4817. However, Werner

Most authorities attribute the discovery of the first cobalt-ammine to an otherwise unknown Citizen Tassaert (no first name given).[3] Some chemists imply or even openly state that Tassaert was the first to prepare and isolate hexaamminecobalt(III) chloride, the parent compound from which all cobalt-ammines may be considered to be derived. Yet, in 1798, in the course of analyzing the cobalt ore of Tunaberg, he merely observed the brownish mahogany color of the hexaamminecobalt(III) ion $[Co(NH_3)_6]^{3+}$, formed when excess aqueous ammonia is added to a solution of cobalt chloride or nitrate, and he failed to follow up his accidental discovery.[4]

During the first half of the nineteenth century there were a few sporadic investigations of cobalt-ammonia compounds, mostly in solution, by Leopold Gmelin (1822), E. M. Dingler (1829), C. Winkelblech (1835), C. Rammelsberg (1839), and W. Beetz (1844).[5] Although the latter two workers actually analyzed their compounds, Rammelsberg's formulas did not appear to correspond to well-defined distinct compounds, and Beetz, whose analytical methods left much to be desired, did not crystallize his compounds. Credit for "the first distinct recognition of the existence of perfectly well defined and crystallized salts of cobalt-ammonia bases"[6] clearly belongs to Frederick Augustus Genth, as we shall now see.

GENTH'S ARTICLE OF 1851

It was in 1847, while he was Bunsen's assistant at Marburg and during Bunsen's absence in Iceland, that Genth obtained his first results on the cobalt-ammines. The story of their discovery was passed by oral tradition from Genth to Edgar Fahs Smith to Thomas P. McCutcheon to Louis C. W. Baker, the latter two of whom were my instructors in general chemistry at the University of Pennsylvania. Genth was demonstrating qualitative analysis to his students. Immediately before vacation, possibly Christmas, he had just removed the precipitated metals of Analytical Group II by filtration from acidic solution. The procedure called for making the solution basic with potassium hydroxide solution before resaturating with hydrogen sulfide. Since the demonstration laboratory had run out of potassium hydroxide, Genth substituted ammonia water ("spirits of hartshorn"), but he had no time to resaturate the basic solution with hydrogen sulfide, so he put it aside. After vacation he found large, beautiful carmine crystals. Careful repetitions of the procedure, with exposure of the ammonia-containing cobalt solutions to air, but with varying conditions, enabled Genth to prepare several different types of crystals.[7] According to Barker, Genth's

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makes frequent reference to Gibbs' later work carried out by Gibbs alone without Genth, e. g., his work on the so-called "Gibbs Triamminkobaltnitrit," $[Co(NH_3)_3(NO_2)_3]$, and croceo salts, *trans*-$[Co(NH_3)_4(NO_2)_2]X$.

[3]In the *Royal Society Catalogue of Scientific Papers*, 1871, *5*:914, Tassaert's initials are given as B. M., but his initials are not given in any of his three papers in *Annales de Chimie*. Antoine-François Fourcroy (*Système des connaissances chimiques,* Paris: Baudouin, 1801, Vol. V, pp. 108–109) identifies Tassaert as "chimiste prussien chargé des travaux du laboratoire de l'école des mines à Paris." I am indebted for this meager information to Professors William A. Smeaton and Maurice P. Crosland and to Mrs. Susan Court. Authorities in Paris were unable to provide any information.

[4]Citoyen Tassaert, "Analyse du Cobalt de Tunaberg, suivie de plusieurs moyens d'obtenir ce métal à l'état de pureté, et de quelques-unes des ses propriétés les plus remarquables," *Annales de Chimie, ou Recueil de Mémoires concernant la Chimie et les Arts qui en dépendent*, 1798, (1) *28*:92–107.

[5]L. Gmelin, *Neues Journal der Chemie und Physik*, (Neue Reihe) 1822, *5*:235; E. M. Dingler, *Kastners Archiv für die gesammte Naturlehre*, 1839, *18*:249; C. Winkelblech, *Liebigs Annalen der Pharmazie*, 1835, *13*:152, 259; C. Rammelsberg, *Poggendorffs Annalen der Physik und Chemie*, 1839, *48*:155; W. Beetz, *Poggendorffs Ann.*, 1844, *61*:489.

[6]Gibbs and Genth, *Researches*, p. 2.

[7]Louis C. W. Baker, letter to John C. Bailar, Jr., July 22, 1970.

results "were freely communicated verbally to others and a suite of the salts obtained were deposited at the time in the laboratory at Giessen."[8]

Before Genth was able to complete his research on and analyses of these interesting new addition compounds, he emigrated to the United States. Thus it was not until January 1851 that he published his results in an obscure, short-lived German-language journal published in Philadelphia.[9] This virtually forgotten "Preliminary Note on Conjugated Cobalt Compounds" opens as follows:

> Some time ago I began an investigation of highly interesting compounds, in which cobaltic oxide (Co_2O_3), conjugated with ammonia, forms two series of salts, similar to the well-known platinum salts of Reiset, Gros, Magnus, etc.[10] Because until now I had no leisure time to complete the investigation, and I do not know how long it will be until I will be able to take it up again, not long ago I communicated to Professor Hermann Kopp in Giessen the results which I had obtained up to that time, requesting him at the same time to have completed by capable hands the work that I had begun. . . . I consider it my duty, regardless of how limited the results may still now be, nevertheless to withhold them no longer from my countrymen because I thereby hope to render a service to science. Once again I wish to point out that the following is only the *beginning* of an investigation (which, however, I will continue as soon as circumstances permit), and it should be regarded only from this viewpoint.

Genth described the preparation and analysis of what he and Gibbs later were to call purpurecobalt chloride (modern, chloropentaamminecobalt(III) chloride, $[Co(NH_3)_5Cl]Cl_2$), and proposed the formula $Co_2O_3,3NH_4,Cl$, which he regarded as "the chlorine compound of a copulated[11] compound $Co_2O_3,3NH_4$, which plays the role of a metal" (see Table 1).[12] He prepared but did not analyze eight other salts of this series as well as the hexachloroplatinate(IV), which he analyzed. He observed that the chloride aquates in boiling aqueous solutions, but he did not investigate the resulting aquopentaamminecobalt(III) or roseocobalt chloride. Genth also mentioned but did not analyze an orange compound ($[Co(NH_3)_6]Cl_3$, or luteocobalt chloride) obtained in preparing the carmine salt. According to Gibbs' and Genth's paper of 1856,

> Though the analyses here given [in Genth's paper of 1851] were from necessity not sufficiently complete and extended to fix the constitution of the bases in question, yet the

[8]George F. Barker, "Obituary Notice: Frederick Augustus Genth," *Proceedings of the American Philosophical Society*, 1901, *40*:x–xxii, p. xiv, and "Memoir of Frederick Augustus Genth," *Biographical Memoirs, National Academy of Sciences*, 1902, *4*:201–231. Inquiries directed to Professors F. Kröhnke (Justus-Liebig-Universität Giessen), K. Dimroth (Philipps-Universität Marburg), and K. Freudenberg (Universität Heidelberg) revealed that Genth's compounds have not been preserved at any of these three universities.

[9]Friedr. Aug. Genth, "Vorläufige Notiz über gepaarte Kobaltverbindungen," *Nordamerikanischer Monatsbericht für Natur- und Heilkunde,* 1851, *2*:8–12, abstracted in *Zentralblatt,* 1851, 417. Genth's results, along with those of Fremy and Claudet, are also discussed in the review "Ueber ammoniakalische Kobaltverbindungen," *Liebigs Ann. Chem.*, 1851, *80*:275–281.

[10]At that time the ammines of platinum, e.g., Reiset's first chloride, $[Pt(NH_3)_4]Cl_2$; Reiset's second chloride, *trans*-$[Pt(NH_3)_2Cl_2]$; Gros' salt, *trans*-$[Pt(NH_3)_4Cl_2]Cl_2$; and Magnus' green salt, $[Pt(NH_3)_4]$-$[PtCl^4]$, were well-known, analyzed compounds. Compared to these, the ammines of cobalt, however, had been little investigated.

[11]The term *copula* had originally been used by Charles Gerhardt (*Annales de chimie et de physique*, 1839, [2] *72*:184) to denote the organic part of a molecule that might be joined to an inorganic part. In 1841 Jöns Jacob Berzelius (*Jahresbericht*, 1841, *21*:108) applied these ideas of *copulated* or *conjugated* compounds to coordination compounds in what was one of the earliest systematic attempts to explain their constitution.

[12]In these "pre-Karlsruhe" formulas, Co_2 is equivalent to our modern Co; i.e., Co refers to an equivalent weight rather than an atomic weight. Genth called the purpureocobalt copula *cobaltic oxide-ammonium (Kobaltoxyd-Ammonium).*

Table 1. Analytical data (%) and formulas for purpureocobalt chloride (chloropentaamminecobalt(III) chloride)

	Genth (Jan. 1851)[a]	Claudet (Oct. 1851)[a]	Fremy (July 1852)[a]	Gregory (1853)[a]	Gibbs and Genth (July 1856)[a]	Theoretical
Co	24.60[b]	23.60	22.6	—	23.56	25.53
Cl	42.49	42.28	40.1	—	42.43	42.47
N	18.04[c]	27.50	26.2	—	28.01	27.96
H	5.19[c]	6.37	6.4	—	6.11	6.04
NH_3	21.93[c]	—	30.1	—	—	34.00
Empirical formula	$Co_2O_3N_3H_{12}Cl_3$	$Co_2Cl_3N_5H_{16}$	$Co_2Cl_3,5AzH_3,HO$	$Co_2Cl_3N_5H_{15}$	$Co_2Cl_3N_5H_{15}$	$CoCl_3N_5H_{15}$
Copula or radical	$Co_2O_3,3NH_4$	NH^2Co	—	—	$5\overset{\frown}{NH3 \cdot Co_2}$	$[Co(NH_3)_5Cl]^{2+}$
Constitutional formula	$Co_2O_3,3NH_4Cl$	$3(NH_4Cl)+2(NH_2Co)$ (Berzelius) or $Cl_3\begin{cases}NH_2Co_2\\NH_3,NH_4\\NH_3,NH_4\end{cases}$ (Graham) or $ClN\begin{cases}H_2\\Co_2\end{cases} + 2ClN\begin{matrix}H_3\\NH_4\end{matrix}$ (Graham)	$Co_2Cl_3,5AzH_3,HO$	$Co_2Cl_3,5NH_3$	$5\overset{\frown}{NH_3 \cdot Co_2}Cl \cdot Cl_2$	$[Co(NH_3)_5Cl]Cl_2$

[a] Co_2 = modern Co.
[b] Calculated from 33.50% Co_3O_4.
[c] Calculated from 23.23% NH_4.

fact is indisputable that this memoir contained not merely the first announcement of the existence of ammonia-cobalt bases, but also a scarcely less accurate and complete description of two of these bases than any which has since appeared.[13]

FREMY'S NOTICES OF 1851

In two short notices communicated to the French Académie des Sciences published later in 1851, Edmond Fremy briefly described his work on cobalt-ammines. In the first, published April 7, 1851, a mere three months after Genth's notice, Fremy announced that "the oxides of cobalt, more highly oxygenated than the protoxide [the lower oxide, CoO], can unite with ammonia and form new series of salts having ammonia, oxygen, and cobalt for the base." He concluded his first note thus: "It is known that the composition of these oxides of cobalt has not yet been definitely established; by making them enter into definite and crystallized compounds, it will be easy for me to cause the uncertainties that their history still presents to disappear."[14]

In his second notice, published May 26, 1851, four months after Genth's notice, Fremy began:

> Since my researches on cobalt have increased quite a bit and have necessitated numerous analyses, I have thought that, in order to assure myself priority in a work that has already

[13] Gibbs and Genth, *Researches,* p. 2.

[14] E. Fremy, "Recherches sur le cobalt," *Comptes rendus hebdomadaires de l'Académie des Sciences, Paris*, 1851, [3] *32*:509–510, pp. 509, 510. An abridged German translation can be found as "Vorläufige Notiz über das Kobalt von E. Fremy," *J. prakt. Chem.*, 1852, *52*:511.

occupied me for a long time, I should be allowed first to make known the new facts that I have established in a certain manner, while waiting to record next, in a complete work, my analytical results.[15]

He then announced a method for producing "new salts of cobalt in which the metal is united simultaneously to oxygen and to chlorine; the base of these salts is thus a chlorinated oxide of cobalt." He briefly described the carmine chloropentaammines of cobalt and recommended their use in the preparation of chemically pure cobalt. He specifically mentioned that the chlorine in the chloropentaammine ion is not immediately precipitated by silver nitrate,[16] a fact which had a profound influence on Sophus Mads Jørgensen's choice of a lifetime field of research and hence affected the future course of coordination chemistry. As we shall see, Gibbs and Genth overlooked this important fact in their analyses. Unlike Genth, however, Fremy did not give analytical data or suggest formulas in his notes of 1851.

CLAUDET'S ARTICLE OF 1851

In a paper dated August 29, 1851, and published in October 1851, nine months after the publication of Genth's notice and five months after Fremy's second notice, Frédéric Claudet, working at University College, London, described in more detail than Genth the preparation, properties, and analysis of Genth's chloropentaammine-cobalt(III) chloride and its "double salts" with platinic chloride and mercuric chloride. He also prepared seven other salts of the series.[17] Claudet recognized the similarity of these cobalt-ammines to the better-known platinum-ammines of Gros and Reiset, and he proposed for the chloride the composition 3Cl,2Co,5N,16H, which he formulated in three ways—the first according to Berzelius' theory of copulated compounds and the last two according to Thomas Graham's ammonium theory (see Table 1):

(1) $3(NH^4Cl) + 2(NH^2Co)$, that is, "a compound of 3 equivalents of chloride of ammonium with 2 equivalents of an ammonia, in which 1 atom of hydrogen is replaced by cobalt;"[18]

(2)

$$Cl^3 \begin{cases} NH^2Co^2 \\ NH^3,NH^4 \\ NH^3,NH^4 \end{cases}$$

where "NH^2Co^2 represents an ammonium in which 2 equivalents of hydrogen are replaced by 2 equivalents of cobalt; while NH^3,NH^4 represents an ammonium in which 1 equivalent of hydrogen is replaced by ammonium itself, as the hydrogen of ammonia is replaced by ethyle, methyle, & c. in Wurtz and Hofmann's bases;"[19] and

[15] E. Fremy, "Recherches sur le cobalt (suite)," *Compt. rend.*, 1851, [3] *32*:808–810, p. 808.
[16] *Ibid.*, p. 809.
[17] Frédéric Claudet, "On a Class of Ammoniacal Compounds of Cobalt," *The London, Edinburgh and Dublin Philosophical Magazine and Journal of Science*, 1851, (4) *2*:253–260; *Ann. chim. phys.*, 1851, (3) *33*:483–493.
[18] Claudet, "On a Class of Ammoniacal Compounds," *Phil. Mag.*, p. 256, *Ann. chim. phys.*, p. 488.
[19] *Phil. Mag.*, p. 257.

$$\mathrm{ClN}\left\{\begin{matrix}\mathrm{H}^2\\ \\ \mathrm{Co}^2\end{matrix}\right. + 2\mathrm{ClN}\left\{\begin{matrix}\mathrm{H}^3\\ \mathrm{NH}^4\end{matrix}\right. \tag{3}$$

that is, "a double salt, composed of 1 equivalent of a chloride of cobalt-ammonium and 2 equivalents of a chloride of ammonium, in which the fourth atom of hydrogen is replaced by ammonium."[20] Claudet concluded his paper by mentioning Fremy's two notices and Genth's paper, as abstracted in the *Chemical Gazette,* 1851, *9*:286. In a footnote on page 260 he specifically states: "The priority of discovery of this new class of salts belongs, not to M. Fremy, but to Dr. Genth, whose researches were published early in 1850;[21] but unfortunately in a journal, the circulation of which appears to be confined to the German physicians of the United States."

Except for a more detailed discussion of analytical procedures, analytical results, and suggested formulas, Claudet added nothing essentially new to Genth's memoir. His paper does show, however, that in the nineteenth century some works of American chemists, even those published in obscure journals, came to the attention of European chemists without any great time lapse, largely through the medium of abstracts, summaries, and translations.

Meanwhile, Genth was finding difficulty in fulfilling his promise to continue his work on his "gepaarte Kobaltverbindungen." In a letter of October 2, 1851, written from Philadelphia to Justus Liebig in Munich, he wrote:

> I have fallen behind with my conjugated cobalt compounds, and now, when office hours begin, I have packed up all my salts (about 30), which I have prepared and partially investigated in the course of this summer, in order to resume and hopefully complete this work next summer. . . . Meanwhile I still had to prepare further compounds of this series as soon as Herr Fremy also published his investigations, whereby it will become possible for him to come to an understanding with us about the conjugated cobalt compounds within a year's time.[22]

FREMY'S MEMOIR OF 1852

In a note of January 1852 Fremy mentioned both Claudet's and Genth's work, again claimed priority in the preparation of purpureocobalt salts, and again promised that his "redaction définitive" would soon appear.[23] In July of 1852 Fremy's promised extensive (55-page) and often-cited paper finally appeared.[24] Fremy took a broad approach and sought "to establish further fundamental differences between the salts of nickel and those of cobalt."[25] He noted that "ammonia possesses the quite

[20] *Ibid.*

[21] The correct date is January 1851. Otherwise Claudet's assignment of priority to Genth is correct.

[22] "Justus Liebig, Briefwechsel mit Einzelnen," Bayerische Staatsbibliothek, Munich.

[23] E. Fremy, "Note sur les sels ammoniaco-cobaltiques suroxygénés," *Ann. chim. phys.*, 1852, *34*:90–91.

[24] E. Fremy, "Recherches sur le cobalt," *Ann. chim. phys.*, 1852, [3] *35*:257–311; also published in abridged German translations as "Untersuchungen über das Kobalt von E. Fremy," *Liebigs Ann. Chem.*, 1852, *83*:227–249, 289–317, and *J. prakt. Chem.*, 1852, *57*:81–106.

[25] *Ibid.* (*Ann. chim. phys.*), p. 258.

remarkable property of causing a superoxidation [*suroxydation*] of cobalt which is not produced by any other method."[26] He recognized that the reactions undergone by ammoniacal solutions of cobalt salts on exposure to atmospheric oxygen are very complicated and yield a number of different products. He described the preparation, properties, and analysis of the nitrates, sulfates, and chlorides of five series of cobalt-ammines, for which he also proposed formulas. It was in this paper that Fremy also originated the system of naming coordination compounds according to their colors, a system widely used for many years, even after Werner proposed his own system of nomenclature in 1897, and still occasionally used today.

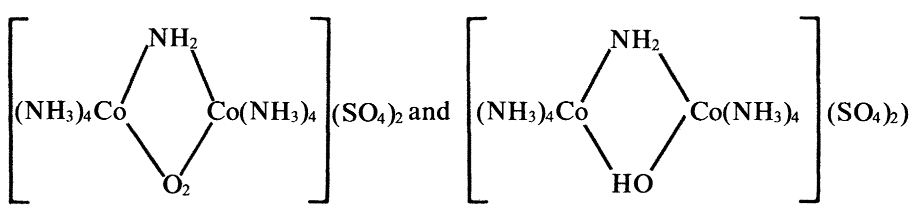
Fremy divided his five series into two broad classes: *sels ammoniacobaltiques,* which consists of one series of pink compounds of divalent cobalt (CoO), previously prepared by Heinrich Rose, and *sels ammoniacobaltiques suroxygénés,* divided into four "new" series, all based on trivalent or tetravalent cobalt,[27] for which he also proposed the shorter designation *cobaltiaque* (*Kobaltiak,* in German):

(1) *Sels ammoniacobaltiques*
$CoCl,3AzH^3,3HO$ (modern, $[Co(NH_3)_6]Cl_2$)

(2) *Sels d'oxycobaltiaque* (olive brown)
$2(CoO^2,SO^3),5AzH^3,3HO$ (modern, $[(NH_3)_5Co\text{-}O_2\text{-}Co(NH_3)_5](SO_4)_2\cdot 3H_2O$)

(3) *Sels de lutéocobaltiaque* (yellow-orange)
$Co^2Cl^3,6AzH^3$ (modern, $[Co(NH_3)_6]Cl_3$)

(4) *Sels de fuscobaltiaque* (insoluble, brown)
$Co^2O^3,2SO^3,4AzH^3,4HO$ (not a single compound, possibly a mixture of



(5) *Sels de roséocobaltiaque* (red or pink)
$Co^2Cl^3,5AzH^3,HO$ (modern, $[Co(NH_3)_5H_2O]Cl_3$, but the compound prepared by Fremy was actually $[Co(NH_3)_5Cl]Cl_2$)

Fremy noted the similarity in stability between his luteocobalt and roseocobalt salts and the better-known platinum-ammines of Gros, Reiset, Peyrone, Raewsky, and Gerhardt. Although he proposed formulas for his five series, he promised to

[26] *Ibid.*, p. 262.
[27] These compounds, the first polynuclear coordination compounds to be discovered, do not contain cobalt(IV). For a discussion of the problem of so-called tetravalent cobalt see George B. Kauffman, "Alfred Werner's Research on Polynuclear Coordination Compounds," *Coordination Chemistry Reviews,* 1973, *9*:339–363, pp. 347–349.

consider in a future article the question of the constitution of his cobalt compounds and to relate them to Gerhardt's platinum-ammines, but he does not appear to have done so.

Two of Fremy's five series are of direct interest to us here in connection with Genth's and Gibbs' work—luteocobalt and roseocobalt salts. For the first, which were mentioned in passing by Genth in 1851 as "an orange-colored cobalt salt" without analytical results or a formula, Fremy presented analytical data and a formula, which agreed with modern formulations. With regard to Fremy's so-called *sels de roséocobaltiaque,* it should be noted that he included under this designation both the salts known today as roseocobalt—that is, aquopentaammines, $[Co(NH_3)_5H_2O]X_3$—as well as the salts known today as purpureocobalt, for example, chloropentaammines, $[Co(NH_3)_5Cl]Cl_2$—that is, the "carmine-red salt" first prepared and analyzed by Genth, whose preparation and analysis was repeated shortly thereafter by Claudet. The properties of and analytical data for Fremy's *chlorhydrate de roséocobaltiaque* (Table 1) clearly show it to be identical with Genth's and Claudet's compound. As we have mentioned, this compound is extremely important in the history of the cobalt-ammines. In Fremy's case, as well as in Jørgensen's, as we shall see, it provided the impetus for his investigation:

> A young chemist working in my laboratory, Monsieur [P.P.] Deherain, called my attention to a pink salt which was formed in a solution of ammoniacobaltic chloride exposed to the air for several days and then redissolved in excess hydrochloric acid.... In order to explain its formation, it was necessary to undertake the researches, the results of which I am making known in this memoir.[28]

In their memoir of 1856, Gibbs and Genth, in claiming for Genth the priority in the discovery of purpureocobalt and luteocobalt salts, a priority to which he is clearly entitled, state that "Fremy appears not to have been aware that these two bases had been described in a manner little less complete than his own two years before the appearance of his memoir."[29] This statement is accepted uncritically and repeated by Barker in his biographical studies of Genth.[30] Yet Fremy specifically stated that his *chlorhydrate de roséocobaltiaque* "had already been obtained by several chemists, and described in a precise manner by Monsieur Aug. Genth. It had likewise been studied very carefully by Monsieur Claudet."[31] Nevertheless, Fremy did describe his last four series of compounds as "new," which may have accounted for Gibbs' and Genth's incorrect statement.

ROGOJSKI'S AND GREGORY'S ARTICLES

Earlier in the same year that Fremy's complete memoir appeared, J. B. Rogojski, who was working in Charles Gerhardt's laboratory, announced (February 2, 1852) to the French Académie des Sciences his discovery of a new compound different from Claudet's and supposedly different from any of Fremy's compounds.[32] Rogojski naturally cited only Fremy's two preliminary notices in *Comptes rendus,* which gave

[28] Fremy, "Recherches sur le cobalt," p. 301.
[29] Gibbs and Genth, *Researches,* p. 3.
[30] Barker, "Obituary Notice," p. xiv.
[31] Fremy, "Recherches sur le cobalt," p. 302.
[32] [no initial] Rogojski, "Sur de nouvelles combinaisons du cobalt," *Compt. rend.*, 1852, *34*:186–187; summarized in *J. prakt. Chem.*, 1852, *55*:357.

no details or analytical data, so he was unaware that the yellow-orange compound that he was describing was identical with Fremy's *chlorhydrate de lutéocobaltiaque,* that is, $[Co(NH_3)_6]Cl_3$. Rogojski unknowingly confirmed Fremy's analysis (which had not yet appeared) for this compound, giving the formula $Co^2Cl^3,3N^2H^6$, or ClH,N^2H^5co, where co or *cobalticum* = 2/3 Co. In analogy with $N^2H^4Pt^2$, *diplatinamine,* he gave the name *dicobaltinamine* to the base N^2H^5co, that is, one double molecule of ammonia in which one hydrogen atom has been replaced by its equivalent of *cobalticum.* Rogojski's complete paper did not appear until August 1854, two full years after Fremy's memoir of 1852.[33] Here he gave detailed preparations, analytical data, and formulas for the *chlorhydrate,* chloroplatinate, chlorosulfate, and nitrate of his *dicobaltinamine* base. He mentioned Fremy's article of 1852 but said nothing regarding the identity of his salts with Fremy's *sels de lutéocobaltiaque.* He formulated his chloride (*chlorhydrate*) as ClH,N^2H^5co, according to the ammonia theory, and as

$$Cl,N\begin{cases}NH^4\\H^2\\co\end{cases}$$ according to the ammonium theory.

Meanwhile in 1853, William Gregory, Professor of Chemistry at Edinburgh and a former student of Liebig's, wrote a two-paragraph letter to Liebig agreeing with Claudet's formula for $[Co(NH_3)_5Cl]Cl_2$ rather than with Fremy's and confirming the composition of $[Co(NH_3)_6]Cl_3$ (see Table 1).[34]

GIBBS' AND GENTH'S MEMOIR OF 1856

Thus, as we have just seen, Genth's short "preliminary notice" of 1851 attracted the attention of a number of other workers to the cobalt-ammines, resulting in an increased research activity in this field. On July 21, 1852, Gibbs, who had followed Genth's directions and prepared Genth's bases in his laboratory in New York, wrote him:

> I enclose you herewith a small quantity of my orange-cobalt compound [hexaamminecobalt(III) or luteocobalt]. . . . Please let me have your opinion of it. I think it identical with yours. Let me urge you to go on with your investigation, as it must lead to very interesting results independently of the beauty of the compounds in question.[35]

Genth's response was apparently prompt, for in a second letter dated July 26, 1852, Gibbs wrote:

> In reply to your proposition I can only say that I will willingly join you in your investigation, provided that on your return to Philadelphia you find that your engage-

[33] Rogojski, "Sur de nouvelles combinaisons du cobalt," *Ann. chim. phys.*, 1854, [3] *41*:445–460; translated into German as "Ueber einige neue Kobaltverbindungen von J. B. Rogojski," *J. prakt. Chem.*, 1854, *56*:491–504.

[34] "Vermischte Notizen von Dr. W. Gregory: Kobaltsalze," *Liebigs Ann. Chem.*, 1853, *87*:125; summarized in *ibid.*, p. 188.

[35] This and the following quotation from Gibbs' letters are given by Barker ("Obituary Notice," p. xv) without ascription of the source. Despite numerous inquiries, I have been unable to locate any pertinent correspondence between Gibbs and Genth.

> ments will prevent you from accomplishing your work alone. You ought, if possible, to have the entire credit which is justly due to you. If, however, you cannot undertake the matter alone, then I will add my labors to yours and we will publish in our joint names.

Thus began the collaboration of the two men on an investigation which deserves much more recognition in the annals of coordination chemistry than it has received.

Several months later, in November of 1852, Gibbs made his first original contribution to this field by discovering a new cobalt-ammonia base, which he obtained by passing oxides of nitrogen into solutions of Genth's compounds. The salts, now known as nitropentaamminecobalt(III), or xanthocobalt compounds, $[Co(NH_3)_5NO_2]X_2$, possess a dark sherry-wine or yellow-brown color, and the new base, that is, cation, differed from those previously described in that it contained the nitrite group as well as ammonia and cobalt. Gibbs communicated his results to the American Association for the Advancement of Science at its Cleveland meeting in August 1853, but he did not publish them.

It was not until 1856, however, that Gibbs' and Genth's joint results appeared in their lengthy and seminal monograph *Researches on the Ammonia-Cobalt Bases.*[36] Several years had been required to complete the research, for the analytical portion of the work was both difficult and protracted. Gibbs and Genth described in detail the preparation, properties, analytical data, and reactions of thirty-five salts of four cobalt-ammine bases: (1) Fremy's roseocobalt or aquopentaamminecobalt(III), (2) Genth's purpureocobalt or chloropentaamminecobalt(III) (Table 1),[37] (3) Genth's luteocobalt or hexaamminecobalt(III), and (4) Gibbs' xanthocobalt or nitropentaamminecobalt(III). In eleven cases they reported crystallographic data obtained by the famous American mineralogist James D. Dana. They substantially adopted Fremy's color system of nomenclature but modified it for the English language by dropping the suffix *iaque*.

For the first time the closely related roseo and purpureo compounds were clearly differentiated, although Gibbs and Genth erred in supposing them to be isomeric, an understandable error in view of the great similarity in composition. They showed that Fremy's chloride of *roséocobaltiaque* was the chloride of the purpureocobalt series. Gibbs and Genth used the characteristic dichromates, oxalates, and hexachloroplatinates(IV) to distinguish the two series, but they failed to recognize that roseo salts contained a molecule of water coordinated or, to use the terminology of the time, conjugated to the cobalt atom, or that purpureo salts contained an atom of chlorine conjugated to the metal atom.

Sophus Mads Jørgensen (1837–1914) dominated coordination chemistry from 1878 until Werner began his systematic studies in 1893.[38] Jørgensen's epoch-making endeavors on the metal-ammine complexes emerged from an also apparently unrelated minor investigation, "On the Detection of Strychnine in Urine,"[39] in which the crystallization of an alkaloid polyiodide was first reported. Jørgensen next wanted to prepare similar compounds, substituting inorganic bases for the alkaloids. He chose the compound purpureocobalt chloride, first prepared, as we have seen, in 1851 by

[36]See n. 1 above.

[37]Barker ("Obituary Notice," p. 210) incorrectly attributed the discovery of roseocobalt and purpureocobalt salts to Genth and Claudet, respectively.

[38]George B. Kauffman, "Sophus Mads Jørgensen and the Werner-Jørgensen Controversy," *Chymia*, 1960, *6*:180–204.

[39]S. M. Jørgensen, "Om Paavisning af Strychnin i Urinen," *Tids. Phys. Chem.*, 1866, *5*:1–8.

Genth and subsequently investigated by Gibbs and Genth, who reported that treatment of purpureocobalt chloride with cold concentrated sulfuric acid yielded an apparently chlorine-free acid sulfate whose aqueous solution gave no precipitate with silver nitrate. Gibbs and Genth analyzed their acid sulfate for cobalt, sulfuric acid, hydrogen, and nitrogen only.[40] Jørgensen confirmed these observations and used the resulting acid sulfate to prepare his desired polyiodosulfate. On analyzing his product, however, and finding a discrepancy between his results and the expected values, he investigated more thoroughly and found that the substance contained not only iodine, but chlorine as well, since the purpureocobalt acid sulfate still contained one atom of chlorine for every atom of cobalt. Jørgensen was so intrigued by the fact that treatment of purpureocobalt chloride with cold concentrated sulfuric acid removes only two of the three chlorine atoms while the remaining third chlorine is not precipitated by silver nitrate that he devoted the rest of his career almost exclusively to the investigation of metal-ammine complexes. When one considers that Werner's coordination theory, which forms the basis for today's coordination chemistry, was based largely upon the experimental data carefully and painstakingly accumulated over a number of years by Jørgensen, the importance of Genth's purpureocobalt chloride and Gibbs' and Genth's purpureocobalt acid sulfate becomes abundantly clear.

Gibbs and Genth provided an elaborate nine-page theoretical discussion of the advantages and disadvantages of the different systems for formulating the constitution of their compounds, which they compared to the known platinum-ammines. They noted that roseocobalt and luteocobalt are triacid bases (tripositive cations) and that purpureocobalt and xanthocobalt are diacid bases (dipositive cations). Their formulas for the conjugate radicals,[41] based upon Berzelius' dualistic theory and employing the connecting circumflex as a symbol of conjugation, as suggested by Kolbe, may be compared with the modern formulas according to Werner as follows (Co equals an equivalent of the metal rather than an atom):

Color name	**Gibbs & Genth**	**Werner**	**Modern name**
Roseocobalt	$5NH_3 \cdot Co_2$	$[Co(NH_3)_5H_2O]^{3+}$	Aquopentaamminecobalt(III)
Purpureocobalt	$5NH_3 \cdot Co_2$	$[Co(NH_3)_5Cl]^{2+}$	Chloropentaamminecobalt(III)
Luteocobalt	$6NH_3 \cdot Co_2$	$[Co(NH_3)_6]^{3+}$	Hexaamminecobalt(III)
Xanthocobalt	$NO_2 \cdot 5NH_3 \cdot Co_2$	$[Co(NH_3)_5NO_2]^{2+}$	Nitropentaamminecobalt(III)

CONCLUSION

Thus, almost forty years before Werner's coordination theory, a long series of salts of each of these four bases had been described in detail, and Fremy, Genth, Gibbs, and others had recognized that some atoms within the compound formed a more intimately bonded group than others. These data were later more satisfactorily accounted for by the Blomstrand-Jørgensen chain theory's concept of "nearer" and "farther" atoms and the Werner theory's concept of *Hauptvalenz* and *Nebenvalenz*.

[40] Gibbs and Genth, *Researches*, p. 31.
[41] *Ibid.*, p. 61.

Gibbs and Genth pointed the way to the future by correctly predicting coordination compounds in which "one or more equivalents of ammonia" are replaced by "an equal number of equivalents of a compound ammonia, as for example, methylamin, ethylamin, &c." and those in which cobalt could be replaced by other metals.[42] They concluded their elaborate and extended memoir, which George F. Barker claimed "has always ranked among the highest chemical investigations ever made in this country [U.S.A.],"[43] with the statement, "we invite the attention of chemists to a class of salts which for beauty of form and color, and for abstract theoretical interest, are almost unequalled either among organic or inorganic compounds."[44] As we have seen, their invitation was accepted by a number of chemists, particularly by Sophus Mads Jørgensen beginning in 1878 and Alfred Werner beginning in 1893. Frederick Augustus Genth thus deserves credit not only for "the first distinct recognition of the existence of perfectly well defined and crystallized salts of ammonia-cobalt bases"[45] but also for producing with Wolcott Gibbs a memoir that attracted chemists to a field which is now experiencing a renaissance of activity.

[42]*Ibid.*, p. 65.
[43]"Obituary Notice," p. xvi.
[44]Gibbs and Genth, *Researches*, p. 67.
[45]*Ibid.*, p. 2.